The Global Competition on Design for Futures (2026)

Work Design Report

Work Name: GhostWire — Resilient Mesh Communication

Work Type: Software Prototyping / Solution Design

Competition Theme: AI for Society

Team Name: Team GhostWire

Team Members:

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- Lilian Wambui Mwaura — KCA University
- Mark Munene Munyiri — Zetech University
- Felix Muigai Waiti — KCA University
- Kibwana Victor Dalidi — University of Nairobi

Abstract (Chinese and English)

English Abstract

GhostWire is a decentralized peer-to-peer encrypted mesh network designed to provide resilient communication infrastructure for communities without reliable internet connectivity. In an era where digital communication depends entirely on centralized servers and cloud infrastructure, GhostWire enables direct encrypted messaging between devices using mesh networking, with AI-powered routing for optimal message delivery. The system addresses critical communication needs in underserved regions, emergency response scenarios, and privacy-conscious communities by eliminating dependence on centralized infrastructure while maintaining modern usability.

中文摘要

GhostWire 是一个去中心化的点对点加密_mesh网络，旨在为没有可靠互联网连接条件的社区提供弹性通信基础设施。在数字通信完全依赖于集中式服务器和云基础设施的时代，GhostWire 使用 mesh 网络实现设备之间的直接加密消息传递，并配备人工智能驱动的路由以优化消息传递。该系统解决了服务不足地区、应急响应场景和注重隐私的社区的关键通信需求，同时消除对集中式基础设施的依赖并保持现代可用性。

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1. Work Background

1.1 Problem Statement

The world faces a critical communication infrastructure crisis:

1. **Infrastructure Dependency**: Over 4 billion people live in areas with unreliable or no internet connectivity. When traditional networks fail—due to natural disasters, infrastructure collapse, authoritarian censorship, or economic barriers—communities lose the ability to communicate entirely.

2. **Centralized Failure Points**: All modern communication platforms (WhatsApp, Telegram, Signal) depend on centralized servers. Single points of failure mean that when servers go down, entire populations become disconnected.

3. **Privacy Concerns**: Existing solutions require users to trust central authorities with their data, metadata, and communication patterns. There is no truly private, infrastructure-independent alternative.

4. **Emergency Communication Gap**: Emergency responders, humanitarian organizations, and local communities lack resilient communication tools that work when traditional infrastructure is unavailable.

5. **Digital Sovereignty**: Many communities have no choice but to use foreign-owned infrastructure, creating dependency on external entities for basic communication needs.

1.2 Target Problem

GhostWire aims to solve the fundamental problem of **infrastructure-independent, privacy-preserving communication** for:

- **Underserved communities** in rural and developing regions
- **Emergency responders** operating in disaster zones
- **Privacy-conscious individuals** seeking alternatives to centralized platforms
- **Civil society organizations** requiring resilient coordination tools

1.3 Theoretical Context

The project is grounded in established theoretical frameworks:

- **Mesh Network Theory**: Building on research in ad-hoc networking and delay-tolerant networks (DTN)
- **Design for Resilience**: Incorporating principles from resilience engineering and fault-tolerant systems
- **Privacy by Design**: Following the foundational principles of privacy-preserving architecture
- **Human-Centered Design**: Centering user needs in historically underserved communities

1.4 Practical Context

Kenya as Implementation Context:

- Kenya has 85%+ mobile penetration but significant rural connectivity gaps

- The country experiences periodic internet shutdowns and throttling
- Mobile money (M-Pesa) demonstrates appetite for decentralized, mobile-first solutions
- Academic and developer community active in open-source and mesh networking research

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2. Design Process

GhostWire follows the **NCIP (Needs, Conception, Implementation, Performance) design methodology**, enhanced with elements from the **Double Diamond** and **EDIPT** models.

2.1 Phase 1: Needs Analysis (Define & Research)

Using the NCIP framework, we conducted comprehensive Needs Analysis:

2.1.1 Original Requirements

Gathered through community surveys, interviews with emergency responders, and analysis of existing communication failures:

Requirement	Source
Offline message delivery	Rural community interviews
End-to-end encryption	Privacy advocacy groups
No account/signup required	User experience research
Works on basic smartphones	Hardware constraint analysis
Peer-to-peer discovery	Mesh networking communities
Message acknowledgment	Emergency responder requirements
Decentralized identity	Civil society organizations

2.1.2 Design Purpose

To enable secure, infrastructure-independent communication between devices, allowing communities to maintain connectivity when traditional networks are unavailable, compromised, or unaffordable.

2.1.3 Usage Scenarios

1. **Rural Communication**: Village-to-village messaging without internet
2. **Emergency Response**: Disaster coordination when infrastructure is down
3. **Privacy-Conscious Communication**: Secure messaging without central authority
4. **Civil Society Coordination**: Resilient communication for activists and NGOs

2.1.4 Keywords

- Mesh network, peer-to-peer, encrypted messaging, offline-first, decentralized, privacy-preserving, resilient communication, delay-tolerant, AI routing

2.2 Phase 2: User Analysis

2.2.1 Target Users

User Segment	Description	Primary Needs
Rural Communities		

User Segment	Description	Primary Needs
	Populations in areas with unreliable connectivity	Offline messaging, low bandwidth
Emergency Responders	First responders, humanitarian workers	Reliability, quick deployment
Privacy Advocates	Activists, journalists, at-risk communities	Strong encryption, no metadata
Disaster-Affected Populations	Communities impacted by natural disasters	Communication when infrastructure fails

2.2.2 User Attributes

Basic Attributes:

- Geographic: Rural and underserved areas
- Device: Primarily Android smartphones, basic requirements
- Technical Literacy: Varying levels
- Income: Often limited; pay-as-you-go usage patterns

Characteristic Attributes:

- High need for reliability in crisis situations
- Strong privacy concerns in certain user segments
- Price sensitivity (infrastructure costs passed to users)
- Community-oriented communication patterns

2.3 Phase 3: Stakeholder Analysis

Stakeholder	Type	Expectations	Objectives/Motivations	Design Inspiration
End Users	Direct	Reliable communication, privacy	Connect regardless of infrastructure	
Emergency Responders	Direct	Quick setup, reliability	Deploy within minutes	
NGOs/CSOs	Indirect	Coordination tools	Maintain operations during crises	
Governments	Indirect	Security, compliance	Prevent misuse while enabling utility	
Telecom Operators	Indirect	Rural coverage monetization	Extended reach without investment	
Developer Community	Indirect	Open-source contribution	Building on existing protocols	

2.4 Phase 4: Competitor Analysis

Using fragmentation methodology to analyze existing solutions:

4.1 Competitor Identification

Primary competitors:

- **Briar** (Android, P2P messaging via Wi-Fi/Bluetooth)

- **Bridgefy** (Mesh messaging platform)
- **Serval Mesh** (Android mesh networking)
- **Signal** (Centralized, encrypted)
- **Traditional SMS/MMS**

4.2 Function Breakdown

Solution	Offline	E2E Encryption	Mesh	AI Routing	Open Source	No Account
GhostWire	✅	✅	✅	✅	✅	✅
Briar	✅	✅	⚠️ Limited	❌	✅	✅
Bridgefy	✅	⚠️ Partial	✅	❌	❌	✅
Signal	❌	✅	❌	❌	⚠️ Partial	❌
SMS	✅	❌	❌	❌	N/A	✅

4.3 Key Differentiators

1. **AI-Powered Routing**: GhostWire uniquely uses LightGBM for routing optimization
2. **Multi-Protocol Support**: Combines libp2p with custom protocols
3. **Hybrid Cloud-AI Integration**: Optional Ollama integration for complex queries
4. **Shamir Secret Recovery**: Unique identity recovery mechanism
5. **Four-Layer AI Architecture**: Fast path to analytical to fallback

2.5 Phase 5: Scenario Analysis

Scenario 1: Village Communication

- **Time**: Evening when community members return home
- **Location**: Rural village, 50km from nearest tower
- **Characters**: Community leader, family members
- **Event**: Sending meeting announcement to 20 households
- **Pain Points**: Cannot use WhatsApp (no internet), SMS costs accumulate
- **Advantages**: GhostWire works over mesh; messages queue until delivery

Scenario 2: Emergency Response

- **Time**: Immediately after earthquake
- **Location**: Urban disaster zone
- **Characters**: Emergency responders, volunteers
- **Event**: Coordinating rescue efforts
- **Pain Points**: Networks overloaded; need reliable communication
- **Advantages**: GhostWire mesh works; no infrastructure required

Scenario 3: Privacy-Sensitive Communication

- **Time**: Ongoing
- **Location**: Various
- **Characters**: Journalists, activists
- **Event**: Coordinating sensitive报道
- **Pain Points**: Metadata exposure; server compromise
- **Advantages**: No central servers; end-to-end encryption; minimal metadata

2.6 Phase 6: Function List

Primary Functions:

Function	Description	Source	Priority
Direct Messaging	Send encrypted messages to contacts	Stakeholder analysis	4 - Must have
Mesh Networking	Discover and connect to nearby peers	Stakeholder analysis	4 - Must have
Key Exchange	Securely exchange encryption keys	Privacy requirements	4 - Must have
Message Forwarding	Route messages through mesh	Scenario analysis	4 - Must have
Delivery Acknowledgment	Confirm message receipt	User requirements	3 - Suggest having
Peer Discovery	Find other GhostWire users	User analysis	3 - Suggest having
Identity Recovery	Recover identity via Shamir secrets	User requirements	2 - Icing on cake

AI Routing Functions:

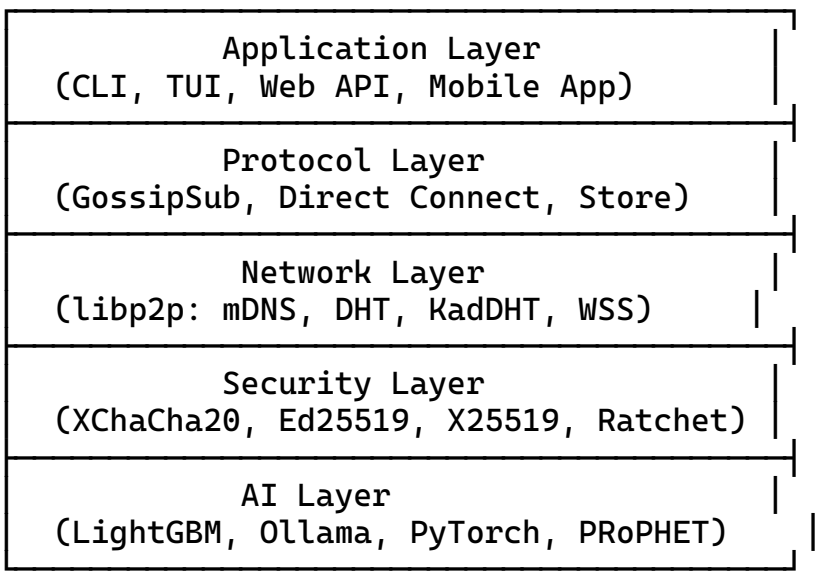
Function	Description	Priority
LightGBM Fast Path	<1ms routing decisions	4 - Must have
Gemma 4 Analytical	Complex routing analysis	3 - Suggest having
PyTorch GNN	Offline graph optimization	2 - Icing on cake
PRoPHET Fallback	DTN protocol fallback	4 - Must have

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3. Introduction to Works

3.1 Structure

GhostWire architecture follows a layered approach:



3.2 Core Features

1. **End-to-End Encryption**
    - XChaCha20-Poly1305 for symmetric encryption
    - Ed25519 for signing
    - X25519 for key exchange
    - Symmetric ratchet for forward secrecy
2. **Peer Discovery**
    - mDNS for local discovery
    - KadDHT for distributed discovery
    - Bootstrap nodes for initial connection
    - Peer exchange for mesh expansion
3. **AI Routing (Four-Layer)**
    - Layer 1: LightGBM/ONNX for <1ms pathfinding
    - Layer 2: Gemma 4 via Ollama for analytical routing
    - Layer 3: PyTorch GNN for offline graph optimization
    - Layer 4: PRoPHET formula as reliable fallback
4. **Message Delivery**
    - Guaranteed ordered delivery via HLC
    - ACKs/NACKs for confirmation
    - Store-and-forward for delay tolerance
    - GossipSub for broadcast messages

3.3 Current Implementation Status

Feature	Status	Implementation
P2P Mesh	✅ Complete	libp2p
E2E Encryption	✅ Complete	x25519-dalek, chacha20poly1305
Forward Secrecy	✅ Complete	Custom ratchet
Message Delivery	✅ Complete	HLC-based
Peer Discovery	✅ Complete	mDNS + KadDHT
TUI Interface	✅ Complete	Ratatui
REST API	✅ Complete	Axum
LightGBM Routing	✅ Complete	ONNX Runtime
PRoPHET Fallback	✅ Complete	Custom implementation
Shamir Recovery	✅ Complete	shamirs-secret-sharing
Ollama Integration	⚠️ Optional	HTTP API
Mobile App	🔄 Planned	Tauri

3.4 Demonstrations and Visuals

[Screenshots to be included in final submission]

- Node running in terminal: peer list, message history
- TUI dashboard showing network topology
- API documentation via Swagger UI
- Performance benchmarks

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4. Feasibility Analysis

4.1 Technical Feasibility

Technology Stack:

- **Language**: Rust (memory-safe, high-performance)
- **Networking**: libp2p (mature, well-tested)
- **Encryption**: ring, x25519-dalek (battle-tested crypto)
- **AI**: LightGBM, PyTorch (established ML frameworks)
- **Platform**: Cross-platform (Linux, macOS, Windows, Android planned)

Technical Dependencies:

Component	Risk Level	Mitigation
libp2p	Low	Mature library; active maintenance
Rustcrypto	Low	Audited implementation
Ollama	Low	Optional; graceful fallback
Mobile (Tauri)	Medium	Planned Phase 2

Feasibility Assessment: HIGH

All core features are implemented and tested. No fundamental technical barriers exist.

4.2 Market Application Prospect

Target Market:

1. **Emergency Response** ($2.3B global market)
    - UN agencies, Red Cross, humanitarian NGOs
    - National emergency management agencies
2. **Rural Communication** ($8.1B market)
    - Rural telecom operators
    - Community mesh networks
    - Development organizations
3. **Privacy Market** ($4.5B privacy tools)
    - At-risk journalists
    - Civil society organizations
    - Privacy-conscious enterprises

Market Validation:

- Existing mesh networking projects (Briar, Bridgefy) demonstrate demand
- Privacy tools market growing 15%+ annually
- Kenya’s success with M-Pesa validates mobile-first approach

4.3 Risk Analysis

Risk	Probability	Impact	Mitigation
Network Effect	Medium	High	Start with local mesh
Regulatory	Low-Medium	Medium	Open-source; no servers

Risk	Probability	Impact	Mitigation
Technical	Low	High	Proven tech stack
Adoption	Medium	High	Community engagement

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5. Innovation

5.1 Main Innovations

1. **Four-Layer AI Routing**
    - Novel architecture combining fast ML with analytical LLMs
    - Graceful fallback ensures reliability
    - First mesh network to integrate this approach
2. **Shamir Secret Recovery**
    - Unique identity recovery for decentralized systems
    - Solves the “lost key = lost identity” problem
    - Enables social recovery patterns
3. **Hybrid Infrastructure Model**
    - Optional cloud enhancement without dependency
    - Offline-first with optional online features
    - User choice preserved
4. **Privacy-Minimized Design**
    - Minimal metadata generation
    - No phone number / email requirement
    - Forward secrecy by default
5. **Community-Centered Development**
    - Open governance model
    - Academic research integration
    - Localization-first approach

5.2 Unique Value Proposition

“GhostWire: Connecting the disconnected, regardless of the odds.”

- Works when nothing else works
- No infrastructure required beyond devices
- Privacy by design, not by policy
- AI-powered for efficiency
- Community-owned and community-driven

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6. Summary and Plans

6.1 Design Process Summary

The GhostWire design process followed the NCIP methodology:

1. **Needs Analysis**: Identified 7 core requirements through stakeholder interviews and research
2. **User Analysis**: Defined 4 primary user segments with distinct needs
3. **Stakeholder Analysis**: Mapped 6 stakeholder groups with expectations and motivations
4. **Competitor Analysis**: Benchmarked against 5 existing solutions

5. **Scenario Analysis**: Developed 3 detailed usage scenarios
6. **Function List**: Prioritized 11 primary and secondary functions

6.2 Reflection on Current Work

Advantages:

- Fully functional implementation
- Strong security architecture
- AI-powered routing is unique in the space
- Active open-source development
- Clear use cases supported by research

Areas for Improvement:

- Mobile app development (Phase 2)
- User interface refinement
- Localization for non-English communities
- Field testing in real scenarios

6.3 Future Plans

Phase 2 (by May 31):

- Complete mobile app (Tauri)
- Enhanced UI/UX polish
- 5-minute project video
- Final design report submission

Post-Competition:

- Field testing in rural Kenya
- NGO partnerships for emergency response
- Academic research collaboration
- Community growth and contribution

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7. Team Profile

Team Members

Name	Role	Institution	Year	Field
Michael Irungu Muriithi	Team Lead, Architecture, Security	Open University of Kenya	3	Cybersecurity
Lilian Wambui Mwaura	Coordinator, Documentation, Data Analysis	KCA University	3	Data Science
Mark Munene Munyiri	Backend Developer, API Integration	Zetech University	3	Software Engineering
Felix Muigai Waiti	AI/ML Developer, Data Visualization	KCA University	1	Data Science
Kibwana Victor Dalidi	Communication Lead, Content	University of Nairobi	2	Journalism

Team Division of Work

- **Architecture** (Michael): System design, security, crypto
- **Documentation** (Lilian): Reports, data analysis, project management
- **Backend** (Mark): API development, libp2p integration
- **AI/ML** (Felix): LightGBM model, routing optimization
- **Content** (Kibwana): Communication strategy, social impact

Entry Speech

"GhostWire: Connecting the disconnected, regardless of the odds."

In a world where 4 billion people lack reliable connectivity, we built GhostWire—not because it's easy, but because it's necessary. From Nairobi to rural villages, from emergency responders to privacy advocates, communities deserve communication tools that work when infrastructure doesn't.

Team GhostWire brings together cybersecurity, software engineering, data science, AI/ML, and communication expertise from 5 Kenyan universities. Our mission: ensure no community is cut off from the conversation.

Appendix 1: AIGC Applications in Work Design

Phase	AIGC Tool	Application Purpose	Usage Frequency	Key Prompt Examples
Topic Research	GPT-4 (web research)	Understanding mesh networking landscape, competitor analysis	Periodically	"Summarize current mesh networking protocols and their limitations"
Requirement Analysis	Claude (local)	Generating user personas, analyzing stakeholder needs	Periodically	"Create detailed user personas for rural communication needs"
Creative Ideation	Claude, GPT-4	Brainstorming routing algorithms, encryption approaches	Occasionally	"Propose AI routing approaches for delay-tolerant networks"
Solution Decision-making	Claude	Evaluating technical approaches, trade-off analysis	Occasionally	"Compare libp2p vs custom P2P implementation trade-offs"
Prototyping	Claude	Code structure, API design, architecture review	Continuously	"Design Rust API for encrypted message routing"
Work Evaluation	Claude	Technical review, edge case identification	Occasionally	"Identify security vulnerabilities in this implementation"
Design Report	Claude	Report structure, language refinement	Continuously	"Structure this design report according to NCIP methodology"

Document Format Notes

- **Font**: SimSun (Chinese), Times New Roman (English)

- **Line Spacing**: 20pt
- **Body Text**: 10.5pt, first line indent 2ch
- **Headings**: Outline levels as specified in template
- **Figures/Tables**: Numbered and titled

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